

Respondent-Information Arms: Manuscript Comparisons

Luna and Muse across five prompt arms, replicating the manuscript’s comparison families

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Scope and method

The primary analysis covers **Luna** and **Muse** at N = 500, the two models carried furthest in this extension. A supplementary section repeats the same comparisons for a wider set of five models at N = 100, where every model has complete five-arm coverage but far fewer respondents.

Each cohort evaluates every model on the **same** first-N respondents of the frozen seed-42 sample, across five prompt arms:

Arm	Information supplied
D	demographic profile only
D+I	profile plus the target-omitted ideal-point score
A only	the other 29 observed answers, no demographics
D+A	profile plus the other 29 answers
D+A+I	profile, other answers, and the score

Arm	Information supplied
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Comparison families follow the manuscript implementations in `Code/Boris/`: individual accuracy and PRE (`compare plus party-pre.R`), aggregate accuracy (`calculate aggregate accuracy.R`), regression weights (`regression weights.R`), rare profiles (`rare profiles.R`), and joint ideal-point fits (`evaluate-pilot-local.R`).

Two baselines recur. **PRE** is the proportional reduction in error against the pooled binary-majority rule. **Party PRE** is against each respondent’s own party’s majority on that question, estimated from the full 17,949-respondent human source rather than from the cohort.

Table 1: Cohorts analysed

Cohort	Respondents	Models	Arms	Answers
Luna and Muse, N = 500	500	2	5	150000
Broader model comparison, N = 100	100	5	5	75000

Luna and Muse, N = 500

Individual accuracy

Proportional reduction in error

Proportional reduction in error against the pooled binary-majority rule: the share of the majority rule’s errors that the model avoids. Zero means no better than always predicting the more common answer.

Model	D	A only	D+A	D+I	D+A+I
Luna	0.347	0.445	0.455	0.372	0.444
Muse	0.316	0.466	0.478	0.349	0.473

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 19% and **D+A** closes 20%. On the measure’s own scale that is **A only** 37% better and **D+A** 41% better. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Party PRE

The same reduction, but against each respondent’s **own party’s** majority on that question, estimated from the full 17,949-respondent human source. A negative value means the model does worse than simply applying the party stereotype.

Model	D	A only	D+A	D+I	D+A+I
Luna	0.028	0.174	0.189	0.065	0.172
Muse	-0.017	0.206	0.224	0.031	0.216

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 19% and **D+A** closes 20%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Raw accuracy

Share of individual answers matching the respondent’s real answer. Reported for completeness; it rewards predicting the common answer and so is less informative than PRE.

Model	D	A only	D+A	D+I	D+A+I
Luna	0.725	0.766	0.77	0.735	0.766
Muse	0.712	0.775	0.78	0.726	0.778

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 19% and **D+A** closes 20%. On the measure’s own scale that is **A only** 7% better and **D+A** 8% better. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

By party subgroup

Party PRE within each party

The same reduction, but against each respondent’s **own party’s** majority on that question, estimated from the full 17,949-respondent human source. A negative value means the model does worse than simply applying the party stereotype.

Model	party	D	A only	D+A	D+I	D+A+I
Luna	Democrat	-0.016	0.077	0.095	-0.023	0.045
Luna	Independent	0.112	0.315	0.322	0.193	0.318
Luna	Republican	-0.031	0.103	0.122	-0.002	0.127
Muse	Democrat	-0.096	0.067	0.109	-0.127	0.094
Muse	Independent	0.081	0.334	0.335	0.190	0.331
Muse	Republican	-0.055	0.195	0.209	0.002	0.203

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 18% and **D+A** closes 20%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Aggregate accuracy

Mean absolute error in support proportion

Aggregate accuracy: the mean absolute gap between the model’s and the survey’s support proportion, averaged over the 30 issues. This is a population-level measure, so individual errors in opposite directions cancel.

Model	D	A only	D+A	D+I	D+A+I
Luna	0.1181	0.0991	0.0897	0.1219	0.0968
Muse	0.0830	0.0914	0.0783	0.0942	0.0825

D+A performs best and **D+I** worst. Swapping demographics for the other answers improves it slightly. Of the gap to a perfect score left open by demographics alone, **A only** closes 5% and **D+A** closes 16%. Adding the ideal-point score moves it moderately on top of demographics alone and slightly on top of demographics-plus-answers.

Majority direction correct (% of 30 issues)

Share of the 30 issues where the model gets the majority direction right, regardless of the size of the gap.

Model	D	A only	D+A	D+I	D+A+I
Luna	86.7	86.7	90.0	86.7	80
Muse	86.7	76.7	83.3	83.3	80

Model	D	A only	D+A	D+I	D+A+I
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D performs best and **D+A+I** worst. Swapping demographics for the other answers worsens it substantially. Of the gap to a perfect score left open by demographics alone, **A only** widens it 38% and **D+A** closes 0%. On the measure’s own scale that is **A only** 6% worse. Adding the ideal-point score moves it moderately on top of demographics alone and substantially on top of demographics-plus-answers.

Signed aggregate bias

Signed aggregate gap: positive means the model overstates support across issues, negative means it understates. Zero is unbiased in level, which is compatible with large offsetting errors.

Model	D	A only	D+A	D+I	D+A+I
Luna	0.0165	-0.0098	-0.0191	0.0196	-0.0247
Muse	-0.0509	-0.0370	-0.0314	-0.0681	-0.0375

D is closest to zero and **D+A+I** departs from it most. Swapping demographics for the other answers worsens it moderately. Of the gap to a perfect score left open by demographics alone, **A only** widens it 36% and **D+A** widens it 47%. Adding the ideal-point score moves it moderately on top of demographics alone and moderately on top of demographics-plus-answers.

By party subgroup

Aggregate mean error within each party

Aggregate accuracy: the mean absolute gap between the model’s and the survey’s support proportion, averaged over the 30 issues. This is a population-level measure, so individual errors in opposite directions cancel.

Model	party	D	A only	D+A	D+I	D+A+I
Luna	Democrat	0.1699	0.0914	0.1203	0.1685	0.1235
Luna	Independent	0.1898	0.1201	0.1126	0.1770	0.1190
Luna	Republican	0.2091	0.1480	0.1859	0.2199	0.1811
Muse	Democrat	0.1425	0.1035	0.0969	0.1343	0.0945
Muse	Independent	0.1046	0.1104	0.0940	0.1013	0.0998
Muse	Republican	0.2977	0.1184	0.1247	0.2732	0.1268

A only performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 38% and **D+A** closes 34%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Ideal points

Ideal-point mean absolute error

Mean absolute error between each respondent’s simulated and real ideal point, from a joint human/LLM fit. This asks whether the model places a person correctly on the latent scale, not whether it gets individual items right.

Model	D	A only	D+A	D+I	D+A+I
Luna	0.776	0.281	0.422	0.682	0.421
Muse	0.862	0.248	0.296	0.801	0.320

A only performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 68% and **D+A** closes 56%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Correlation with the human ideal point

Correlation between simulated and real ideal points across respondents: whether the model gets the ordering of people right, ignoring scale and offset.

Model	D	A only	D+A	D+I	D+A+I
Luna	0.719	0.943	0.905	0.824	0.918
Muse	0.715	0.966	0.955	0.867	0.956

A only performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 84% and **D+A** closes 75%. On the measure’s own scale that is **A only** 33% better and **D+A** 30% better. Adding the ideal-point score moves it moderately on top of demographics alone and hardly at all on top of demographics-plus-answers.

Mean signed ideal-point error

Signed ideal-point error. Negative means the model places respondents further left than they are; positive, further right. Invisible to accuracy and PRE.

Model	D	A only	D+A	D+I	D+A+I
Luna	-0.402	-0.117	-0.160	-0.381	-0.167
Muse	-0.017	-0.069	-0.065	-0.005	-0.046

A only is closest to zero and **D** departs from it most. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 56% and **D+A** closes 47%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Polarization ratio

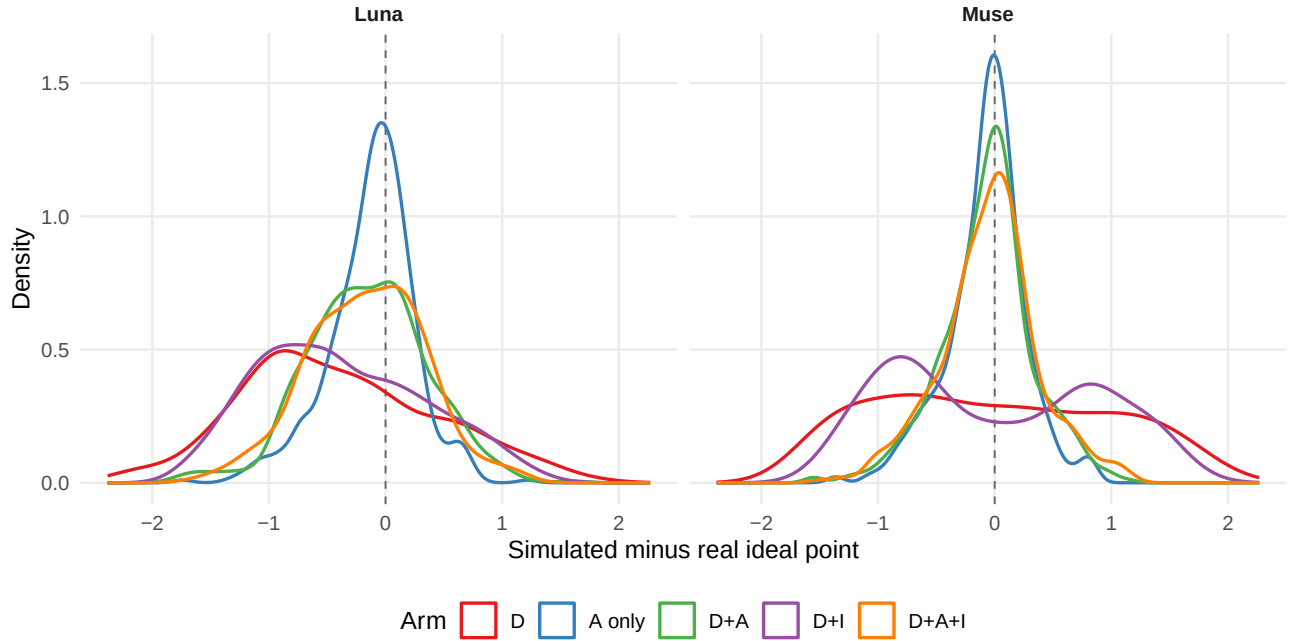
Polarization ratio: the simulated Democrat-Republican median gap divided by the real one. Values above 1 mean the model exaggerates the partisan divide; 1.0 would reproduce it.

Model	D	A only	D+A	D+I	D+A+I
Luna	2.77	1.3	1.74	2.66	1.72
Muse	4.56	1.3	1.49	4.26	1.52

A only comes closest to the reference value and **D** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 89% and **D+A** closes 77%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Deviation densities

Distribution of each respondent’s simulated minus real ideal point, by arm. A curve centred away from the dashed line places people systematically to one side; a tall narrow curve means the model compresses distinct people toward a single position.



Regression weights

Weight differences by predictor: Luna

Regression-weight difference per predictor: $(\text{mean}|\text{LLM coefficient}| - \text{mean}|\text{CCES coefficient}|) / \text{mean}|\text{CCES coefficient}|$. Zero means the model weights that predictor as real respondents do; +1.0 means twice the weight.

Model	predictor	D	A only	D+A	D+I	D+A+I
Luna	age	-0.50	-0.17	-0.22	-0.49	-0.29
Luna	gender	-0.54	0.01	-0.14	-0.30	-0.17
Luna	ideology	0.68	0.07	0.40	0.64	0.50
Luna	married_dummy	0.20	-0.24	0.29	0.56	0.48
Luna	pid_text	0.68	0.22	0.41	0.78	0.34
Luna	race_black	-0.38	-0.06	-0.11	-0.25	-0.04
Luna	race_latino	-0.15	0.01	-0.12	-0.27	-0.21
Luna	registered_dummy	0.20	-0.06	0.13	0.30	0.08

A only comes closest to the reference value and **D+I** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 75% and **D+A** closes 45%. Adding the ideal-point score moves it slightly on top of demographics alone and slightly on top of demographics-plus-answers.

Weight differences by predictor: Muse

Regression-weight difference per predictor: $(\text{mean}|\text{LLM coefficient}| - \text{mean}|\text{CCES coefficient}|) / \text{mean}|\text{CCES coefficient}|$. Zero means the model weights that predictor as real respondents do; +1.0 means twice the weight.

Model	predictor	D	A only	D+A	D+I	D+A+I
Muse	age	-0.51	0.06	0.05	-0.36	-0.02
Muse	gender	-0.63	0.03	-0.11	-0.28	-0.06

Model	predictor	D	A only	D+A	D+I	D+A+I
Muse	ideology	1.63	0.09	0.30	1.00	0.21
Muse	married_dummy	0.54	0.31	0.15	0.07	-0.02
Muse	pid_text	0.36	0.12	0.18	0.69	0.24
Muse	race_black	-0.42	0.11	-0.10	-0.26	0.03
Muse	race_latino	-0.14	-0.09	0.07	-0.16	0.02
Muse	registered_dummy	-0.19	-0.05	0.06	0.41	-0.09

D+A+I comes closest to the reference value and **D** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 81% and **D+A** closes 77%. Adding the ideal-point score moves it moderately on top of demographics alone and slightly on top of demographics-plus-answers.

Rare profiles

Cells are race by party with at least 20 respondents.

Accuracy by profile

Share of individual answers matching the respondent’s real answer. Reported for completeness; it rewards predicting the common answer and so is less informative than PRE.

Model	cell	people	D	A only	D+A	D+I	D+A+I
Luna	Black or African-American Democrat	41	0.693	0.724	0.729	0.690	0.715
Luna	Hispanic or Latino Democrat	30	0.742	0.751	0.764	0.741	0.749
Luna	Hispanic or Latino Independent	23	0.664	0.764	0.759	0.671	0.771
Luna	White Democrat	137	0.823	0.842	0.844	0.824	0.836
Luna	White Independent	102	0.696	0.769	0.779	0.735	0.776
Luna	White Republican	114	0.678	0.724	0.728	0.683	0.729
Muse	Black or African-American Democrat	41	0.671	0.722	0.723	0.671	0.729
Muse	Hispanic or Latino Democrat	30	0.726	0.748	0.761	0.694	0.754
Muse	Hispanic or Latino Independent	23	0.678	0.751	0.762	0.713	0.757
Muse	White Democrat	137	0.808	0.839	0.852	0.805	0.845
Muse	White Independent	102	0.691	0.776	0.777	0.732	0.778
Muse	White Republican	114	0.672	0.754	0.757	0.691	0.754

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 18% and **D+A** closes 20%. On the measure’s own scale that is **A only** 7% better and **D+A** 8% better. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

PRE against the group’s own majority

Error reduction against **this group’s own** majority answer. Negative means the model is worse than assuming everyone in the group answers alike, which is the sharpest available test of stereotyping.

Model	cell	people	D	A only	D+A	D+I	D+A+I
Luna	Black or African-American Democrat	41	-0.167	-0.050	-0.031	-0.180	-0.087
Luna	Hispanic or Latino Democrat	30	-0.084	-0.047	0.009	-0.089	-0.056
Luna	Hispanic or Latino Independent	23	0.072	0.348	0.336	0.092	0.368
Luna	White Democrat	137	-0.001	0.103	0.116	0.000	0.069
Luna	White Independent	102	0.143	0.347	0.377	0.253	0.370
Luna	White Republican	114	-0.069	0.083	0.096	-0.053	0.099
Muse	Black or African-American Democrat	41	-0.254	-0.059	-0.056	-0.254	-0.031

Model	cell	people	D	A only	D+A	D+I	D+A+I
Muse	Hispanic or Latino Democrat	30	-0.154	-0.061	-0.005	-0.285	-0.033
Muse	Hispanic or Latino Independent	23	0.112	0.312	0.344	0.208	0.328
Muse	White Democrat	137	-0.090	0.088	0.160	-0.108	0.123
Muse	White Independent	102	0.128	0.370	0.371	0.245	0.373
Muse	White Republican	114	-0.089	0.181	0.191	-0.026	0.183

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 17% and **D+A** closes 19%. Adding the ideal-point score moves it slightly on top of demographics alone and slightly on top of demographics-plus-answers.

Between-person SD ratio

Between-person spread the model reproduces within a group, as a fraction of the real spread. Values below 1 mean it makes group members more alike than they are.

Model	cell	people	D	A only	D+A	D+I	D+A+I
Luna	Black or African-American Democrat	41	0.332	0.577	0.392	0.378	0.402
Luna	Hispanic or Latino Democrat	30	0.329	0.690	0.520	0.508	0.661
Luna	Hispanic or Latino Independent	23	0.433	0.607	0.700	0.534	0.443
Luna	White Democrat	137	0.304	0.572	0.479	0.381	0.449
Luna	White Independent	102	0.590	0.733	0.686	0.621	0.588
Luna	White Republican	114	0.562	0.722	0.570	0.449	0.462
Muse	Black or African-American Democrat	41	0.358	0.753	0.694	0.413	0.718
Muse	Hispanic or Latino Democrat	30	0.405	0.913	0.871	0.387	0.808
Muse	Hispanic or Latino Independent	23	0.341	0.830	0.891	0.407	0.887
Muse	White Democrat	137	0.447	0.762	0.678	0.304	0.590
Muse	White Independent	102	0.858	1.002	1.022	0.602	0.965
Muse	White Republican	114	0.310	0.829	0.766	0.330	0.776

A only comes closest to the reference value and **D** is furthest from it. Every arm falls short of it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 55% and **D+A** closes 45%. Adding the ideal-point score moves it hardly at all on top of demographics alone and slightly on top of demographics-plus-answers.

Issue-profile correlation

Correlation between the group's real and simulated issue profile: whether the model knows *which* issues this group supports.

Model	cell	people	D	A only	D+A	D+I	D+A+I
Luna	Black or African-American Democrat	41	0.835	0.793	0.825	0.836	0.843
Luna	Hispanic or Latino Democrat	30	0.853	0.851	0.839	0.857	0.846
Luna	Hispanic or Latino Independent	23	0.576	0.609	0.633	0.543	0.683
Luna	White Democrat	137	0.953	0.954	0.962	0.955	0.959
Luna	White Independent	102	0.807	0.732	0.810	0.817	0.797
Luna	White Republican	114	0.818	0.779	0.845	0.795	0.845
Muse	Black or African-American Democrat	41	0.837	0.778	0.822	0.832	0.813
Muse	Hispanic or Latino Democrat	30	0.819	0.752	0.778	0.803	0.774
Muse	Hispanic or Latino Independent	23	0.655	0.579	0.631	0.630	0.600
Muse	White Democrat	137	0.937	0.932	0.945	0.906	0.937
Muse	White Independent	102	0.585	0.721	0.746	0.643	0.750
Muse	White Republican	114	0.762	0.817	0.849	0.802	0.849

D+A+I performs best and **A only** worst. Swapping demographics for the other answers worsens it moderately. Of the gap to a perfect score left open by demographics alone, **A only** widens it 6% and **D+A** closes 10%. On the measure’s own scale that is **A only** 1% worse and **D+A** 3% better. Adding the ideal-point score makes little difference, either on its own or on top of the answers.

Supplementary: broader model comparison

The section below repeats every comparison for a wider set of models, but on only 100 respondents. It is included for breadth across model families rather than precision: arm rankings there move materially with sample size, as the Luna and Muse series from $N = 25$ to $N = 500$ demonstrates. Treat the primary section above as the estimate and this one as context.

Broader model comparison, $N = 100$

Individual accuracy

Proportional reduction in error

Proportional reduction in error against the pooled binary-majority rule: the share of the majority rule’s errors that the model avoids. Zero means no better than always predicting the more common answer.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	0.229	0.364	0.318	0.220	0.313
Glimmer	0.262	0.368	0.375	0.268	0.396
Luna	0.318	0.433	0.450	0.333	0.450
Muse	0.266	0.445	0.454	0.301	0.446
Phi-4	0.259	0.342	0.385	0.273	0.337

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 17% and **D+A** closes 18%. On the measure’s own scale that is **A only** 46% better and **D+A** 49% better. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Party PRE

The same reduction, but against each respondent’s **own party’s** majority on that question, estimated from the full 17,949-respondent human source. A negative value means the model does worse than simply applying the party stereotype.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	-0.103	0.090	0.024	-0.116	0.017
Glimmer	-0.056	0.096	0.106	-0.047	0.137
Luna	0.024	0.189	0.213	0.045	0.213
Muse	-0.050	0.206	0.219	0.000	0.207
Phi-4	-0.060	0.059	0.120	-0.039	0.052

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 17% and **D+A** closes 18%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Raw accuracy

Share of individual answers matching the respondent’s real answer. Reported for completeness; it rewards predicting the common answer and so is less informative than PRE.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	0.682	0.738	0.719	0.679	0.717
Glimmer	0.696	0.740	0.743	0.698	0.751
Luna	0.719	0.766	0.773	0.725	0.773
Muse	0.698	0.771	0.775	0.712	0.772
Phi-4	0.695	0.729	0.747	0.701	0.727

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 17% and **D+A** closes 18%. On the measure’s own scale that is **A only** 7% better and **D+A** 8% better. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

By party subgroup

Party PRE within each party

The same reduction, but against each respondent’s **own party’s** majority on that question, estimated from the full 17,949-respondent human source. A negative value means the model does worse than simply applying the party stereotype.

Model	party	D	A only	D+A	D+I	D+A+I
GPT-OSS	Democrat	-0.229	-0.112	-0.054	-0.240	-0.101
GPT-OSS	Independent	0.110	0.355	0.244	0.081	0.267
GPT-OSS	Republican	-0.260	-0.057	-0.187	-0.252	-0.195
Glimmer	Democrat	-0.186	-0.074	-0.050	-0.202	-0.050
Glimmer	Independent	0.140	0.369	0.360	0.169	0.369
Glimmer	Republican	-0.183	-0.095	-0.073	-0.179	0.015
Luna	Democrat	-0.070	0.128	0.155	-0.093	0.078
Luna	Independent	0.160	0.358	0.378	0.203	0.413
Luna	Republican	-0.061	0.027	0.053	-0.027	0.084
Muse	Democrat	-0.182	0.089	0.097	-0.213	0.085
Muse	Independent	0.122	0.392	0.401	0.230	0.416
Muse	Republican	-0.145	0.076	0.099	-0.092	0.053
Phi-4	Democrat	-0.167	-0.128	-0.035	-0.174	-0.039
Phi-4	Independent	0.099	0.314	0.337	0.157	0.265
Phi-4	Republican	-0.164	-0.092	-0.011	-0.164	-0.137

D+A performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 16% and **D+A** closes 17%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Aggregate accuracy

Mean absolute error in support proportion

Aggregate accuracy: the mean absolute gap between the model’s and the survey’s support proportion, averaged over the 30 issues. This is a population-level measure, so individual errors in opposite directions cancel.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	0.1523	0.1407	0.1683	0.1513	0.1743
Glimmer	0.1327	0.1443	0.1433	0.1330	0.1353
Luna	0.1090	0.1063	0.0940	0.1230	0.0987

Model	D	A only	D+A	D+I	D+A+I
Muse	0.1003	0.1027	0.0970	0.1073	0.0990
Phi-4	0.1227	0.1557	0.1333	0.1260	0.1570

D performs best and **D+A+I** worst. Swapping demographics for the other answers worsens it substantially. Of the gap to a perfect score left open by demographics alone, **A only** widens it 5% and **D+A** widens it 3%. Adding the ideal-point score moves it moderately on top of demographics alone and moderately on top of demographics-plus-answers.

Majority direction correct (% of 30 issues)

Share of the 30 issues where the model gets the majority direction right, regardless of the size of the gap.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	73.3	80.0	73.3	63.3	73.3
Glimmer	80.0	80.0	80.0	70.0	83.3
Luna	86.7	83.3	86.7	83.3	80.0
Muse	80.0	76.7	86.7	73.3	80.0
Phi-4	73.3	73.3	80.0	63.3	76.7

D+A performs best and **D+I** worst. Swapping demographics for the other answers barely matters. Of the gap to a perfect score left open by demographics alone, **A only** closes 0% and **D+A** closes 12%. On the measure’s own scale that is **D+A** 3% better. Adding the ideal-point score moves it substantially on top of demographics alone and slightly on top of demographics-plus-answers.

Signed aggregate bias

Signed aggregate gap: positive means the model overstates support across issues, negative means it understates. Zero is unbiased in level, which is compatible with large offsetting errors.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	-0.1290	-0.0953	-0.1410	-0.1253	-0.1497
Glimmer	-0.1133	-0.1183	-0.1233	-0.1083	-0.1180
Luna	0.0003	-0.0177	-0.0300	0.0170	-0.0347
Muse	-0.0677	-0.0507	-0.0503	-0.0840	-0.0503
Phi-4	-0.0907	-0.0563	-0.0787	-0.1020	-0.1110

A only is closest to zero and **D+A+I** departs from it most. Swapping demographics for the other answers improves it moderately. Of the gap to a perfect score left open by demographics alone, **A only** closes 15% and **D+A** widens it 6%. Adding the ideal-point score moves it hardly at all on top of demographics alone and moderately on top of demographics-plus-answers.

By party subgroup

Aggregate mean error within each party

Aggregate accuracy: the mean absolute gap between the model’s and the survey’s support proportion, averaged over the 30 issues. This is a population-level measure, so individual errors in opposite directions cancel.

Model	party	D	A only	D+A	D+I	D+A+I
GPT-OSS	Democrat	0.1833	0.1373	0.1397	0.1794	0.1476
GPT-OSS	Independent	0.1762	0.1548	0.2024	0.1762	0.2024
GPT-OSS	Republican	0.3067	0.1789	0.2611	0.3022	0.2656

Model	party	D	A only	D+A	D+I	D+A+I
Glimmer	Democrat	0.1873	0.1389	0.1468	0.1921	0.1484
Glimmer	Independent	0.1738	0.1679	0.1595	0.1524	0.1583
Glimmer	Republican	0.3222	0.2011	0.2278	0.3300	0.2067
Luna	Democrat	0.1587	0.0992	0.1127	0.1635	0.1254
Luna	Independent	0.2250	0.1440	0.1381	0.2071	0.1405
Luna	Republican	0.2000	0.1478	0.1822	0.2144	0.1844
Muse	Democrat	0.1325	0.1087	0.1024	0.1548	0.1048
Muse	Independent	0.1357	0.1417	0.1333	0.1464	0.1393
Muse	Republican	0.2956	0.1467	0.1556	0.2756	0.1644
Phi-4	Democrat	0.2087	0.1770	0.1659	0.2040	0.1698
Phi-4	Independent	0.2048	0.1762	0.1714	0.1738	0.1821
Phi-4	Republican	0.2900	0.1800	0.1922	0.2967	0.2178

A only performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 28% and **D+A** closes 22%. Adding the ideal-point score makes little difference, either on its own or on top of the answers.

Ideal points

Ideal-point mean absolute error

Mean absolute error between each respondent’s simulated and real ideal point, from a joint human/LLM fit. This asks whether the model places a person correctly on the latent scale, not whether it gets individual items right.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	0.871	0.270	0.458	0.879	0.473
Glimmer	0.939	0.311	0.426	0.960	0.451
Luna	0.730	0.283	0.393	0.623	0.412
Muse	0.879	0.292	0.302	0.831	0.331
Phi-4	0.893	0.378	0.469	0.909	0.501

A only performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 64% and **D+A** closes 53%. Adding the ideal-point score makes little difference, either on its own or on top of the answers.

Correlation with the human ideal point

Correlation between simulated and real ideal points across respondents: whether the model gets the ordering of people right, ignoring scale and offset.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	0.718	0.938	0.900	0.751	0.903
Glimmer	0.710	0.942	0.921	0.784	0.912
Luna	0.707	0.938	0.918	0.829	0.918
Muse	0.711	0.954	0.955	0.845	0.946
Phi-4	0.690	0.913	0.903	0.726	0.898

A only performs best and **D** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 78% and **D+A** closes 72%. On the measure’s own scale that is **A only** 32% better and **D+A** 30% better. Adding the ideal-point score moves it moderately on top of demographics alone and hardly at all on top of demographics-plus-answers.

Mean signed ideal-point error

Signed ideal-point error. Negative means the model places respondents further left than they are; positive, further right. Invisible to accuracy and PRE.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	-0.095	0.042	0.052	-0.099	0.062
Glimmer	0.063	0.078	0.039	0.044	0.021
Luna	-0.280	-0.066	-0.099	-0.285	-0.096
Muse	0.146	0.014	0.001	0.128	0.060
Phi-4	-0.183	-0.131	-0.121	-0.111	-0.057

D+A+I is closest to zero and **D** departs from it most. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 82% and **D+A** closes 63%. Adding the ideal-point score moves it hardly at all on top of demographics alone and moderately on top of demographics-plus-answers.

Polarization ratio

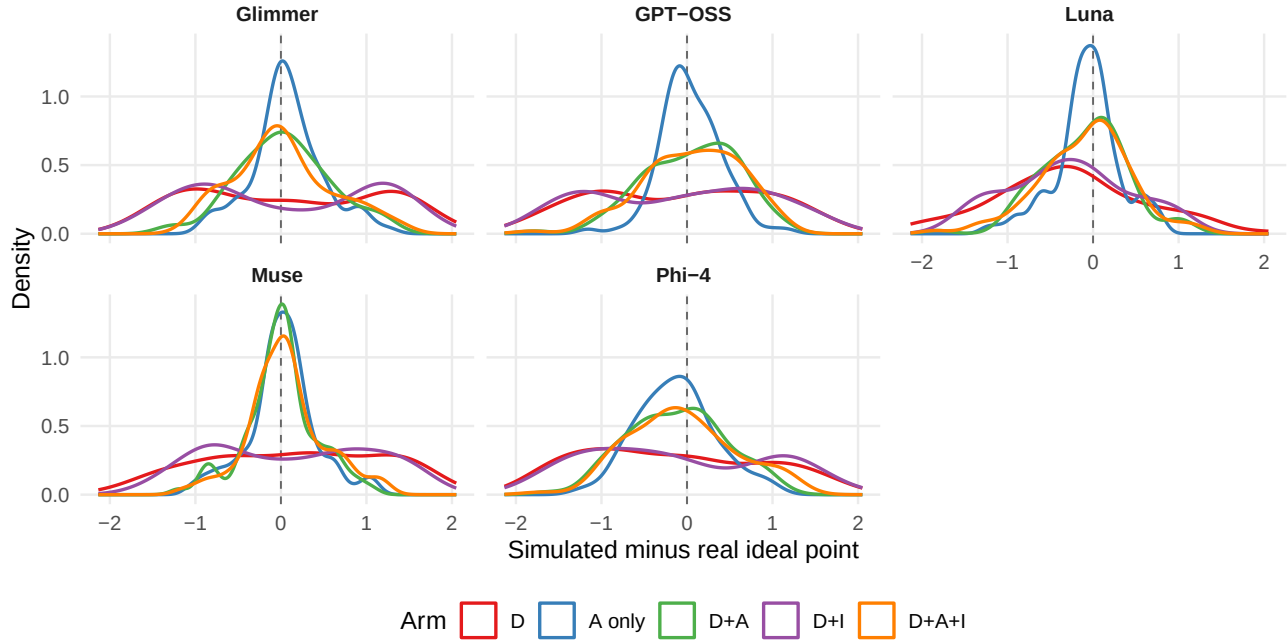
Polarization ratio: the simulated Democrat-Republican median gap divided by the real one. Values above 1 mean the model exaggerates the partisan divide; 1.0 would reproduce it.

Model	D	A only	D+A	D+I	D+A+I
GPT-OSS	4.28	1.16	2.02	4.70	2.00
Glimmer	5.63	1.46	1.81	5.78	2.02
Luna	3.10	1.21	1.64	2.47	1.74
Muse	4.78	1.39	1.44	4.60	1.46
Phi-4	4.45	1.45	2.02	4.98	2.12

A only comes closest to the reference value and **D+I** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 90% and **D+A** closes 77%. Adding the ideal-point score makes little difference, either on its own or on top of the answers.

Deviation densities

Distribution of each respondent’s simulated minus real ideal point, by arm. A curve centred away from the dashed line places people systematically to one side; a tall narrow curve means the model compresses distinct people toward a single position.



Regression weights

Weight differences by predictor: Glimmer

Regression-weight difference per predictor: $(\text{mean}|\text{LLM coefficient}| - \text{mean}|\text{CCES coefficient}|) / \text{mean}|\text{CCES coefficient}|$. Zero means the model weights that predictor as real respondents do; +1.0 means twice the weight.

Model	predictor	D	A only	D+A	D+I	D+A+I
Glimmer	age	-0.75	0.02	0.07	-0.65	0.10
Glimmer	gender	-0.41	-0.13	-0.27	-0.54	-0.34
Glimmer	ideology	1.51	0.16	0.40	1.22	0.35
Glimmer	married_dummy	-0.68	0.04	-0.17	-0.51	-0.14
Glimmer	pid_text	0.75	0.24	0.49	0.96	0.43
Glimmer	race_black	-0.51	0.07	0.18	-0.33	0.11
Glimmer	race_latino	-0.29	-0.25	-0.12	-0.53	-0.22
Glimmer	registered_dummy	-0.74	-0.13	-0.12	-0.68	-0.18

A only comes closest to the reference value and **D** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 82% and **D+A** closes 68%. Adding the ideal-point score makes little difference, either on its own or on top of the answers.

Weight differences by predictor: GPT-OSS

Regression-weight difference per predictor: $(\text{mean}|\text{LLM coefficient}| - \text{mean}|\text{CCES coefficient}|) / \text{mean}|\text{CCES coefficient}|$. Zero means the model weights that predictor as real respondents do; +1.0 means twice the weight.

Model	predictor	D	A only	D+A	D+I	D+A+I
GPT-OSS	age	-0.71	-0.23	-0.30	-0.65	-0.30
GPT-OSS	gender	-0.42	-0.07	-0.30	-0.62	-0.35

Model	predictor	D	A only	D+A	D+I	D+A+I
GPT-OSS	ideology	1.37	0.09	0.58	1.56	0.57
GPT-OSS	married_dummy	-0.44	0.06	-0.16	-0.51	-0.21
GPT-OSS	pid_text	0.68	0.05	0.34	0.50	0.34
GPT-OSS	race_black	-0.46	0.07	0.11	-0.51	-0.11
GPT-OSS	race_latino	-0.47	-0.23	-0.23	-0.52	-0.35
GPT-OSS	registered_dummy	-0.67	-0.02	-0.10	-0.59	-0.36

A only comes closest to the reference value and **D+I** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 84% and **D+A** closes 59%. Adding the ideal-point score moves it hardly at all on top of demographics alone and slightly on top of demographics-plus-answers.

Weight differences by predictor: Luna

Regression-weight difference per predictor: $(\text{mean}|\text{LLM coefficient}| - \text{mean}|\text{CCES coefficient}|) / \text{mean}|\text{CCES coefficient}|$. Zero means the model weights that predictor as real respondents do; +1.0 means twice the weight.

Model	predictor	D	A only	D+A	D+I	D+A+I
Luna	age	-0.25	-0.04	-0.08	-0.19	-0.19
Luna	gender	-0.44	-0.12	-0.44	-0.34	-0.41
Luna	ideology	0.63	0.10	0.42	0.63	0.55
Luna	married_dummy	-0.19	-0.02	-0.14	-0.48	-0.07
Luna	pid_text	0.76	0.13	0.41	0.54	0.39
Luna	race_black	-0.41	-0.07	-0.23	-0.46	-0.11
Luna	race_latino	-0.32	-0.02	-0.13	-0.38	-0.31
Luna	registered_dummy	-0.74	-0.11	-0.28	-0.56	-0.12

A only comes closest to the reference value and **D** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 84% and **D+A** closes 43%. Adding the ideal-point score makes little difference, either on its own or on top of the answers.

Weight differences by predictor: Muse

Regression-weight difference per predictor: $(\text{mean}|\text{LLM coefficient}| - \text{mean}|\text{CCES coefficient}|) / \text{mean}|\text{CCES coefficient}|$. Zero means the model weights that predictor as real respondents do; +1.0 means twice the weight.

Model	predictor	D	A only	D+A	D+I	D+A+I
Muse	age	-0.65	-0.08	0.05	-0.66	-0.09
Muse	gender	-0.54	-0.16	-0.19	-0.61	-0.13
Muse	ideology	1.86	0.13	0.23	1.07	0.25
Muse	married_dummy	-0.39	0.06	-0.05	-0.25	-0.14
Muse	pid_text	0.21	0.12	0.28	0.77	0.33
Muse	race_black	-0.50	0.10	0.15	-0.06	0.06
Muse	race_latino	-0.45	-0.11	-0.03	-0.35	-0.07
Muse	registered_dummy	-0.61	-0.19	-0.12	-0.54	-0.18

A only comes closest to the reference value and **D** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 82% and **D+A** closes 79%. Adding the ideal-point score moves it slightly on top of demographics alone and hardly at all on top of demographics-plus-answers.

Weight differences by predictor: Phi-4

Regression-weight difference per predictor: $(\text{mean}|\text{LLM coefficient}| - \text{mean}|\text{CCES coefficient}|) / \text{mean}|\text{CCES coefficient}|$. Zero means the model weights that predictor as real respondents do; +1.0 means twice the weight.

Model	predictor	D	A only	D+A	D+I	D+A+I
Phi-4	age	-0.53	-0.12	-0.13	-0.54	-0.11
Phi-4	gender	-0.45	-0.35	-0.46	-0.35	-0.56
Phi-4	ideology	0.51	0.33	0.50	0.82	0.57
Phi-4	married_dummy	-0.47	0.03	-0.23	-0.48	-0.31
Phi-4	pid_text	1.35	-0.03	0.48	1.10	0.52
Phi-4	race_black	-0.64	-0.08	-0.18	-0.50	-0.10
Phi-4	race_latino	-0.27	-0.25	-0.19	-0.32	0.08
Phi-4	registered_dummy	-0.69	-0.28	-0.30	-0.63	-0.35

A only comes closest to the reference value and **D** is furthest from it. Every arm overshoots it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 70% and **D+A** closes 50%. Adding the ideal-point score makes little difference, either on its own or on top of the answers.

Rare profiles

Cells are race by party with at least 20 respondents.

Accuracy by profile

Share of individual answers matching the respondent’s real answer. Reported for completeness; it rewards predicting the common answer and so is less informative than PRE.

Model	cell	people	D	A only	D+A	D+I	D+A+I
GPT-OSS	White Democrat	22	0.797	0.797	0.820	0.798	0.811
GPT-OSS	White Republican	23	0.667	0.707	0.674	0.670	0.671
Glimmer	White Democrat	22	0.802	0.814	0.820	0.797	0.821
Glimmer	White Republican	23	0.696	0.700	0.707	0.691	0.733
Luna	White Democrat	22	0.823	0.862	0.859	0.818	0.844
Luna	White Republican	23	0.712	0.739	0.745	0.713	0.746
Muse	White Democrat	22	0.800	0.835	0.850	0.792	0.842
Muse	White Republican	23	0.703	0.752	0.758	0.701	0.735
Phi-4	White Democrat	22	0.803	0.794	0.835	0.802	0.826
Phi-4	White Republican	23	0.699	0.701	0.729	0.699	0.683

D+A performs best and **D+I** worst. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 8% and **D+A** closes 12%. On the measure’s own scale that is **A only** 3% better and **D+A** 4% better. Adding the ideal-point score moves it hardly at all on top of demographics alone and moderately on top of demographics-plus-answers.

PRE against the group’s own majority

Error reduction against **this group’s own** majority answer. Negative means the model is worse than assuming everyone in the group answers alike, which is the sharpest available test of stereotyping.

Model	cell	people	D	A only	D+A	D+I	D+A+I
GPT-OSS	White Democrat	22	-0.252	-0.252	-0.112	-0.243	-0.168

Model	cell	people	D	A only	D+A	D+I	D+A+I
GPT-OSS	White Republican	23	-0.278	-0.122	-0.250	-0.267	-0.261
Glimmer	White Democrat	22	-0.224	-0.150	-0.112	-0.252	-0.103
Glimmer	White Republican	23	-0.167	-0.150	-0.122	-0.183	-0.022
Luna	White Democrat	22	-0.093	0.150	0.131	-0.121	0.037
Luna	White Republican	23	-0.106	0.000	0.022	-0.100	0.028
Muse	White Democrat	22	-0.234	-0.019	0.075	-0.280	0.028
Muse	White Republican	23	-0.139	0.050	0.072	-0.144	-0.017
Phi-4	White Democrat	22	-0.215	-0.271	-0.019	-0.224	-0.075
Phi-4	White Republican	23	-0.156	-0.144	-0.039	-0.156	-0.217

D+A performs best and **D+I** worst. Swapping demographics for the other answers improves it moderately. Of the gap to a perfect score left open by demographics alone, **A only** closes 8% and **D+A** closes 13%. Adding the ideal-point score moves it hardly at all on top of demographics alone and moderately on top of demographics-plus-answers.

Between-person SD ratio

Between-person spread the model reproduces within a group, as a fraction of the real spread. Values below 1 mean it makes group members more alike than they are.

Model	cell	people	D	A only	D+A	D+I	D+A+I
GPT-OSS	White Democrat	22	0.357	0.536	0.693	0.347	0.623
GPT-OSS	White Republican	23	0.271	0.873	0.452	0.302	0.485
Glimmer	White Democrat	22	0.263	0.514	0.638	0.235	0.514
Glimmer	White Republican	23	0.060	0.800	0.633	0.084	0.776
Luna	White Democrat	22	0.425	0.693	0.730	0.531	0.564
Luna	White Republican	23	0.715	0.571	0.517	0.517	0.391
Muse	White Democrat	22	0.521	0.561	0.536	0.339	0.522
Muse	White Republican	23	0.359	0.889	0.715	0.275	0.791
Phi-4	White Democrat	22	0.187	0.434	0.319	0.239	0.702
Phi-4	White Republican	23	0.257	0.468	0.435	0.141	0.578

A only comes closest to the reference value and **D+I** is furthest from it. Every arm falls short of it. Swapping demographics for the other answers improves it substantially. Of the gap to a perfect score left open by demographics alone, **A only** closes 44% and **D+A** closes 34%. Adding the ideal-point score moves it slightly on top of demographics alone and slightly on top of demographics-plus-answers.

Issue-profile correlation

Correlation between the group’s real and simulated issue profile: whether the model knows *which* issues this group supports.

Model	cell	people	D	A only	D+A	D+I	D+A+I
GPT-OSS	White Democrat	22	0.889	0.901	0.915	0.869	0.901
GPT-OSS	White Republican	23	0.690	0.729	0.724	0.720	0.704
Glimmer	White Democrat	22	0.881	0.862	0.897	0.874	0.886
Glimmer	White Republican	23	0.731	0.736	0.774	0.717	0.802
Luna	White Democrat	22	0.945	0.945	0.957	0.955	0.951
Luna	White Republican	23	0.834	0.802	0.836	0.792	0.836
Muse	White Democrat	22	0.930	0.917	0.935	0.896	0.926
Muse	White Republican	23	0.760	0.815	0.828	0.772	0.819
Phi-4	White Democrat	22	0.885	0.812	0.880	0.878	0.879
Phi-4	White Republican	23	0.776	0.757	0.822	0.751	0.742

Model	cell	people	D	A only	D+A	D+I	D+A+I
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D+A performs best and **D+I** worst. Swapping demographics for the other answers worsens it slightly. Of the gap to a perfect score left open by demographics alone, **A only** widens it 3% and **D+A** closes 15%. On the measure’s own scale that is **A only** 1% worse and **D+A** 3% better. Adding the ideal-point score moves it moderately on top of demographics alone and moderately on top of demographics-plus-answers.

Limitations

- The $N = 100$ cohort has 100 respondents per cell, so its rare-profile cells are small and its estimates are correspondingly noisy. The manuscript’s own rare-profile table uses `min_n = 100` per cell, which no extension cohort can currently meet; the threshold here is 20 and every cell reports its count.
- Black Republicans, the manuscript’s sharpest rare-profile case, do not reach the threshold in any cohort.
- Ideal-point fits are single-chain at 10,000 iterations with fixed party starts. They are joint human/LLM fits, so scores are on a common scale within a fit but not across fits.
- Aggregate accuracy is computed on the cohort, while party baselines come from the full human source. That follows the manuscript, but means aggregate and party-PRE numbers rest on different denominators.
- These are exploratory paired comparisons, not significance tests.